Aaron Merchant

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science (BASc), Computer Engineering (Co-op) — *Sept. 2026 – Apr. 2031*

- **Relevant Coursework:** Fundamentals of Programming (C++), Linear Circuits, Linear Algebra, Calculus, Physics of Electrical Engineering

Experience

Embedded Flight Software Developer — Sept. 2026 – Present
Waterloo Aerial Robotics Group (WARG) — *Waterloo, ON*

- Developing **hardware-in-the-loop (HITL)** test infrastructure for **ZeroPilot 4.0**, an open-source **C/C++ FreeRTOS** autopilot on an **STM32H753 (Arm Cortex-M7)**
- Building testbench software that drives flight-controller hardware with simulated inputs to validate firmware before flight
- Working toward automated HIL testing in **GitHub Actions CI** alongside **GoogleTest** unit tests, **SITL** builds, and **Docker**
- Collaborating in a 4-person HITL team through **code reviews** and pull requests, writing test documentation and onboarding new members

Founding Developer — July 2026 – Sept. 2026
Lunate Studios — *Remote*

- Co-founded a game studio and shipped **4 multiplayer games** in **Luau** on a **client-server architecture**
- Engineered **server-side validation**, **RemoteEvents** networking, and **DataStoreService** persistence for player data
- Developed progression, economy, vehicle upgrade, rebirth, quest, and **monetization** systems
- Scaled *Seal A Driveway* to **7.2M+ plays** and **11K+ peak concurrent users**, generating **11M+ Robux** (∼**$58K CAD**) before acquisition by a venture-backed company

Quantum School for Young Students (QSYS) Fellow — Aug. 2025
Institute for Quantum Computing, University of Waterloo — *Waterloo, ON (Remote)*

- Selected for IQC's competitive **quantum computing** enrichment program for high school students
- Studied **quantum algorithms**, **quantum cryptography**, quantum mechanics, quantum optics, entanglement, and experimental physics
- Applied **linear algebra** and complex numbers to model qubit states, quantum gates, and quantum circuits

Projects

Autonomous Targeting Drone | *ArduPilot, Python, OpenCV, MAVLink, F450, Computer Vision*

- Built an autonomous **F450 quadcopter** using ArduPilot and MAVLink for flight control and telemetry
- Implemented **facial recognition and object tracking** with OpenCV for real-time target detection
- Developed **kinematic targeting** and coordinate transformations to align a water actuator with detected targets
- Integrated **PID control, IMU/GPS telemetry**, and autonomous flight routines for stable positioning

Diorama | *Swift, SwiftUI, Metal, C++, GLSL, GitHub Actions*

- Built a native **macOS wallpaper engine** that parses Wallpaper Engine assets and renders 59 scenes from a 114-wallpaper library on Apple Silicon
- Developed a **Metal rendering pipeline** supporting multi-pass effects, particle systems, audio reactivity, and JavaScript-based wallpaper scripts
- Implemented **GLSL-to-Metal shader translation** using SPIR-V, with disk caching and automatic toolchain invalidation
- Optimized rendering to **0.1–0.5% CPU utilization** and built a fully offline SwiftUI app with per-display wallpapers, playlists, and **477 automated tests**

Extrovert AI | *React, Vite, Python, Flask, Tavus API, Supabase, OpenAI*

- Built a full-stack **AI communication platform** using React, Flask, and Tavus for realistic video conversations with AI personas
- Integrated **LLM-powered feedback** to evaluate conversations across six communication traits including clarity, confidence, and adaptability
- Developed **REST APIs** for session analysis, scoring, and persistent user data with Supabase
- Implemented interactive **analytics dashboards** and skill tracking to visualize communication performance across sessions

Technical Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, Swift, SQL, Luau, GLSL
Embedded: STM32, FreeRTOS, ArduPilot, MAVLink, PID Control, IMU/GPS, HITL/SITL Testing
Frameworks & Tools: React, Flask, FastAPI, Node.js, SwiftUI, OpenCV, Metal, Supabase, Git, GitHub Actions, Docker, Linux
Concepts: Embedded Systems, Real-Time Systems, Computer Vision, Client-Server Architecture, REST APIs, CI/CD, Unit Testing